

SkilLine Learning Platform

A full-stack learning management system — video courses that unlock in order, quizzes and graded tasks, paid enrolment, and bootcamp programmes.

Angga Wika Nugraha

Software Engineer · Fullstack Developer

ROLE	TYPE	YEAR
Fullstack Developer	Personal Project	2026

Live demo	skilline-apps.vercel.app
Source code	github.com/AnggaWikaNugraha/fullstack-mevn-LMS
Project page	port-tau-azure.vercel.app/pages/projects/proj_1782661483998
GitHub	@AnggaWikaNugraha

PROJECT OVERVIEW

SkillLine — Learning Management System

An online learning platform where students buy a course, work through video lessons that unlock one at a time, prove what they learned through quizzes and graded tasks, and track their progress — while the platform sells both self-paced courses and scheduled bootcamp programmes.

The visible half is a Vue 3 application. The half that matters most is underneath: a TypeScript backend where every incoming request is validated with Zod, sensitive endpoints are rate limited, and the authentication layer is designed around the things that actually go wrong — shared accounts, leaked tokens, brute-forced codes, and people probing which email addresses exist.

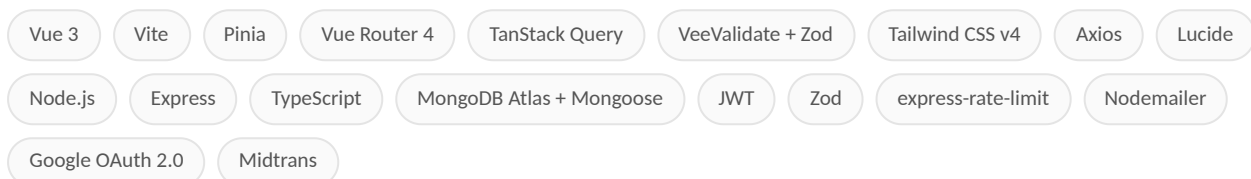
FOR THE STUDENT

- Sign up with a 6-digit email code, or continue with Google
- Browse courses by category, free and paid side by side
- Buy a course through Midtrans; free courses enrol instantly
- Video lessons organised as module → chapter → lesson
- Picks up automatically at the first lesson not yet finished
- Quizzes with a pass mark, and tasks submitted for review
- Progress bar per course, and a purchase history
- Bootcamp programmes with mentors, batches and schedules

BUILT TO HOLD UP

- Email codes expire after 15 minutes, resend locked for 60 seconds
- One active session per account — a second device signs the first out
- Short-lived access tokens, renewed silently in the background
- Password reset signs the account out everywhere
- "Forgot password" never reveals whether an email is registered
- Quiz answers are never sent to the browser
- Lessons unlock in sequence — no skipping ahead
- Every request validated server-side; sensitive routes rate limited

BUILT WITH



Live demo	skilline-apps.vercel.app
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01 Create an account

Sign-up form, with the platform's promise and a student quote on the left.

SkillLine

**Learn anything,
anytime, anywhere.**

Join thousands of students growing
their skills every day.

"This platform completely changed how I learn. The courses are structured
perfectly."

A Alex Johnson
Web Developer

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Create an account
Start your learning journey today

Full Name
John Doe

Email
you@example.com

Password
Min. 8 characters

Confirm Password
Re-enter your password

Create Account

Already have an account? [Sign in](#)

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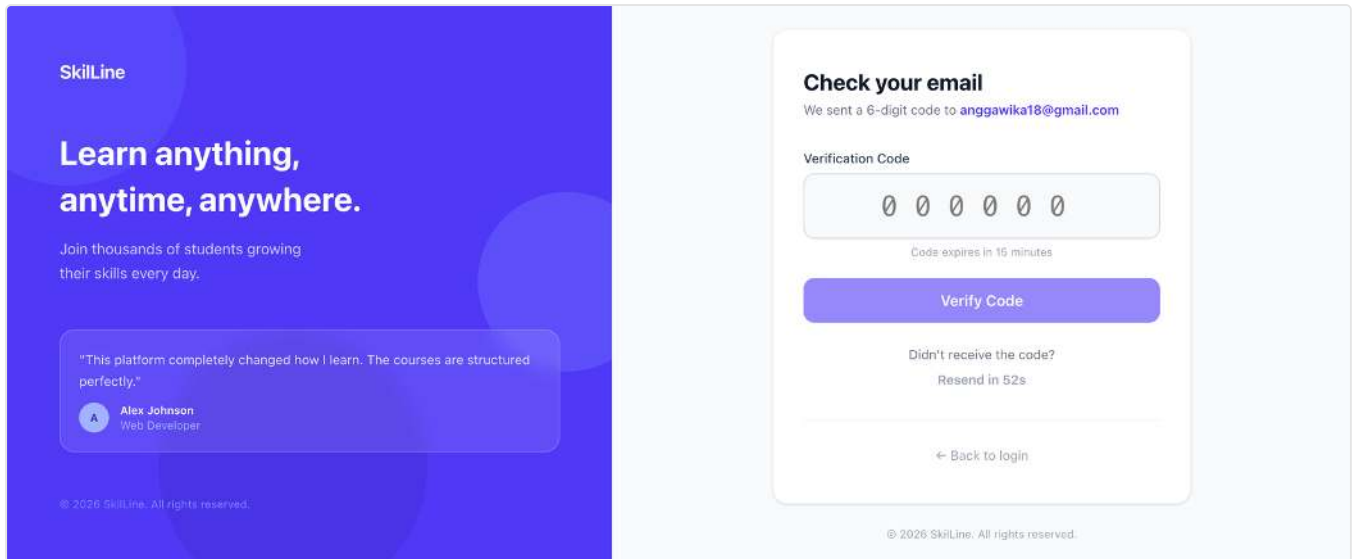
Name, email, password, confirm — with a minimum length enforced on both sides of the wire, not only in the browser.

Two details here are invisible but deliberate. If the email already has an account, the response does not say so outright — it does not hand an attacker a way to test which addresses are registered. And the account is not usable yet: it stays unverified until the code on the next page is entered.

02

Email verification

The 6-digit code screen, with its expiry and resend timer both visible.



A 6-digit code is emailed through Nodemailer and stays valid for **15 minutes**. Resending is locked for **60 seconds**, and the page counts that down rather than leaving the button dead with no explanation.

The two timers do different jobs: the expiry limits how long a leaked code is worth anything, and the cooldown stops the send endpoint being used to flood someone's inbox. Entering a valid code marks the account verified — from then on it can sign in.

03

Sign in

Email and password, with the forgot-password route beside the field that needs it.

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Web Developer

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Welcome back
Sign in to your account to continue

Email
you@example.com

Password [Forgot password?](#)
Enter your password

Sign In

Don't have an account? [Sign up](#)

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Sessions are tied to a device. Signing in on a second device ends the session on the first — one paid account cannot quietly become five people watching the same course. For a platform that sells access, that is the difference between a course being bought and a course being shared.

A successful sign-in issues a **15-minute access token** and a **1-day refresh token**. A short access token limits the damage if one is ever stolen, and the student never feels it: when a request comes back unauthorised the application renews the token in the background and retries — nobody is dropped out of a lesson mid-video. Google sign-in is available as an alternative, and merges into an existing account rather than creating a second one.

04

Forgot password

The recovery form — one field, one button.

The image displays two side-by-side mockups of a web application. The left mockup is a blue landing page for 'SkillLine' with the tagline 'Learn anything, anytime, anywhere.' and a testimonial from Alex Johnson, a Web Developer. The right mockup is a white 'Forgot password?' form. The form includes a title, a subtext 'Enter your email and we'll send you a verification code.', an 'Email' label, a text input field containing 'you@example.com', a blue 'Send Verification Code' button, a 'Remember your password?' link pointing to 'Sign in', and a footer copyright notice '© 2026 SkillLine. All rights reserved.'

This screen always reports success, whether or not the address has an account. That looks like a small thing and is not: a page that says "email not found" is a free tool for checking which of a stolen address list are real customers here. Someone who owns the inbox gets a code; someone fishing gets nothing back.

The same flow doubles as the way in for accounts created through Google. Those accounts start with no password — setting one here means the student can afterwards sign in either way.

05

Set a new password

The reset form, reached only after the emailed code has been verified.

SkillLine

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A Alex Johnson
Web Developer

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Set new password
Must be at least 8 characters long.

New Password
Min. 8 characters

Confirm Password
Re-enter new password

Reset Password

[← Back to login](#)

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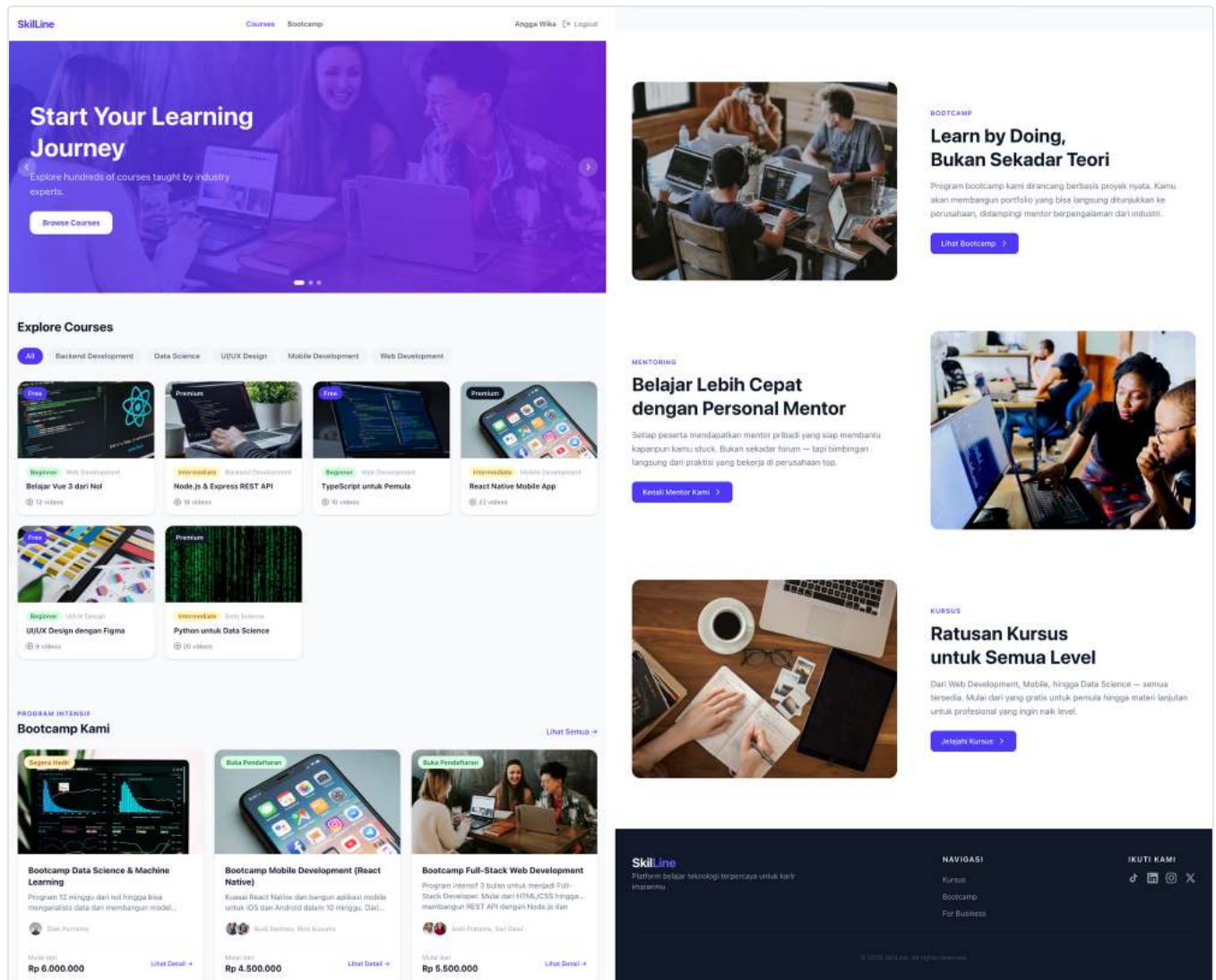
The page cannot be reached by guessing its URL — a valid, unexpired code has to be verified first.

Completing the reset signs the account out of **every** device, including whoever prompted the reset in the first place. If an account was taken over, changing the password is the moment the intruder loses it; leaving their session alive would make the reset pointless.

06

Home page

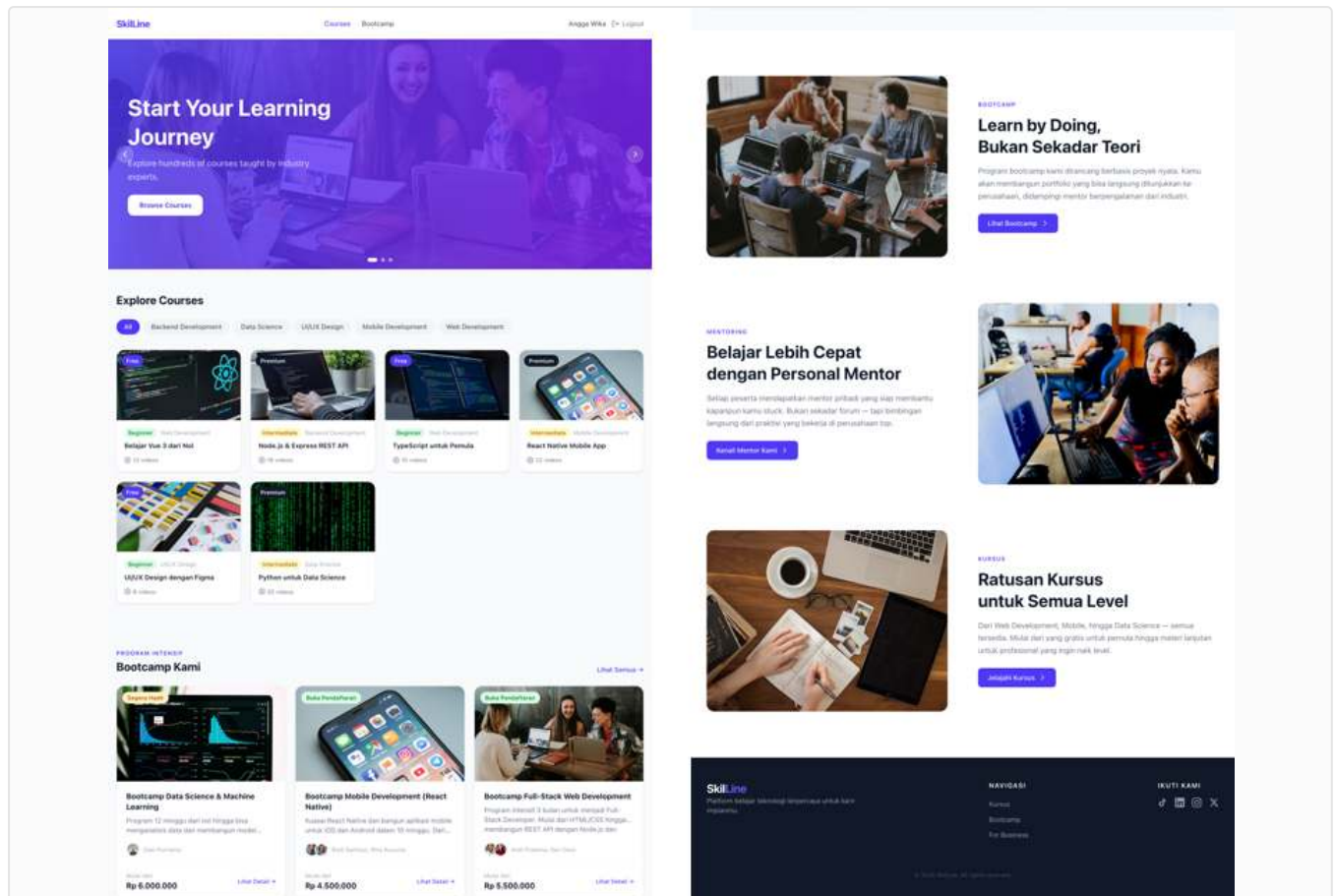
The full page, shown in two columns — top half on the left, continuing on the right.



The landing page has one job: get a visitor into a course. It opens with a rotating hero, drops straight into the course grid with its category filter, then the bootcamp programmes, and closes with the three arguments that sell a learning platform — projects instead of theory, a real mentor instead of a forum, and a catalogue wide enough to matter.

07 On a phone

The same home page at mobile width, again split into two columns.



The layout is not a shrunken desktop page — the grid reflows to a single column, the hero keeps its proportions, and the sections stack in the order that makes sense on a small screen.

Worth its own page because of who is on the other end: most people in Indonesia browse and buy on a phone, and a learning platform that only works properly on a laptop has already lost most of its market before the first lesson plays.

08

Course catalogue

Explore Courses — category pills above, course cards below.

The screenshot displays the SkillLine Course Catalogue interface. At the top, the SkillLine logo is on the left, and 'Courses' and 'Bootcamp' are in the center. A user profile 'Angga Wika' is on the right. Below the header, the 'Explore Courses' section features a row of category pills: 'All', 'Backend Development', 'Data Science', 'UI/UX Design', 'Mobile Development', and 'Web Development'. The main area shows six course cards in a 2x3 grid. Each card includes a difficulty level (Beginner, Intermediate, or Premium), a category, a title, a video count, and a price (or 'Gratis').

Course Title	Difficulty	Category	Videos	Price
Node.js & Express REST API	Intermediate	Backend Development	13	Rp 299.000
TypeScript untuk Pemula	Beginner	Web Development	10	Gratis
React Native Mobile App	Intermediate	Mobile Development	21	Rp 349.000
UI/UX Design dengan Figma	Beginner	UI/UX Design	8	Gratis
Python untuk Data Science	Intermediate	Data Science	18	Rp 399.000
Belajar Vue 3 dari Nol	Beginner	Web Development	12	Gratis

The footer contains the SkillLine logo and tagline 'Platform belajar teknologi terpercaya untuk karir impianmu.' on the left. In the center, under 'NAVIGASI', are links for 'Kursus', 'Bootcamp', and 'For Business'. On the right, under 'IKUTI KAMI', are social media icons for YouTube, LinkedIn, Instagram, and Twitter. A copyright notice '© 2026 Skilline. All rights reserved.' is at the bottom center.

Filtering happens in the browser, so switching category is instant — no spinner between a student and the thing they are looking for.

Every card answers the buying question before it is asked: difficulty level, category, how many videos, and the price — or **Gratis**. Free and paid courses sit in the same grid on purpose. The free ones are the sample that earns the paid ones a chance.

09 Lesson player

The video, with the course curriculum in the right-hand rail — completed, current and locked lessons all visible.

The screenshot shows the SkillLine lesson player interface. The main video area displays a JavaScript crash course for beginners, titled "JavaScript Crash Course Absolute Beginners". The video content includes a code editor with the following code:

```

193
194 // Simple If/Else Statement
195 const x = 30;
196
197 if(x === 10) {
198   console.log('x is 10')
199 } else if(x > 10) {
200   console.log('x is greater
201 } else {
  
```

The video player has a play button and a "Tonton di YouTube" button. The right-hand rail shows the course curriculum, including modules and lessons. The current lesson is "Pengantar JavaScript" (8 menit). The curriculum includes:

- MODUL 1: FUNDAMENTAL JAVASCRIPT
 - Bab 1: Dasar JavaScript
 - Pengantar JavaScript (8 menit)
 - Variabel & Tipe Data (10 menit)
 - Quiz: Dasar JavaScript (Quiz)
 - Bab 2: Function & Array
 - Function & Arrow Function (10 menit)
 - Array Methods Modern (12 menit)
- MODUL 2: VUE 3 CORE
 - Bab 3: Setup & Composition API
 - Instalasi Vue 3 & Vite (8 menit)
 - ref & reactive (10 menit)
 - computed & watch (10 menit)
 - Quiz: Composition API (Quiz)
 - Bab 4: Component & Router
 - Props & Emit (12 menit)
 - Slots & Provide/inject (10 menit)

The bottom of the interface includes a footer with the SkillLine logo, navigation links (Kursus, Bootcamp, For Business), and social media links (Instagram, Facebook, Twitter, YouTube).

A course is structured as **module** → **chapter** → **lesson**, so a long syllabus stays navigable instead of becoming a list of forty videos.

Opening a course does not dump the student at lesson one — it jumps to the first lesson they have not finished, which is where they actually left off. Finishing a lesson unlocks the next one, and the padlocks in the rail show a student how much is ahead of them. Completing the last lesson raises a course-completed confirmation.

10

Quiz

A five-question quiz with its pass mark stated up front, shown in two columns.

SkillLine

Courses Bootcamp

Angga Wika

5 soal · Nilai lulus ≥ 60

1. Apa keuntungan utama menggunakan TypeScript dibanding JavaScript biasa?

A TypeScript berjalan lebih cepat di browser

B TypeScript mendeteksi error tipe data saat kompilasi, sebelum runtime

C TypeScript bisa berjalan di server saja

D TypeScript tidak butuh Node.js

2. Tipe apa yang digunakan untuk variabel yang bisa menyimpan nilai apa saja tapi tetap type-safe?

A any

B unknown

C never

D void

3. Bagaimana cara mendeklarasikan array of string di TypeScript?

A string[]

B Array(string)

C array<string>

D []string

4. Apa itu 'enum' di TypeScript?

A Tipe untuk array angka

B Sekumpulan konstanta bernama yang terkait

C Cara mendefinisikan fungsi

D Alias untuk tipe string

5. Apa arti tipe 'never' di TypeScript?

A Nilai yang bisa null atau undefined

B Tipe untuk fungsi yang tidak mengembalikan nilai

C Tipe yang tidak pernah bisa terjadi — biasanya fungsi yang selalu throw error

D Sama dengan 'void'

Kumpulkan Jawaban

Semua Run

Quiz: Type System

TypeScript untuk Pemula

Progress 2/12 lesson

0 menit

Basic Types & Type Annotation10 menit

Array, Tuple & Enum10 menit

Quiz: Type SystemQuiz

Bab 2: Interface & Function

Interface & Type Alias10 menit

Function Types & Optional Params8 menit

Classes & Access Modifiers10 menit

MODUL 2: TYPESCRIPT LANJUTAN

Bab 3: Generics & Utility Types

SkillLine

Platform belajar teknologi terpercaya untuk karir impianmu.

NAVIGASI

Kursus

Bootcamp

For Business

IKUTI KAMI

[d](#) [in](#) [@](#) [x](#)

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The important detail is not on screen: **the correct answers are never sent to the browser**. The question and its options are; which one is right stays on the server and the grading happens there. Anyone who opens the developer console sees nothing useful — a quiz that ships its own answer key is decoration, not assessment.

The pass mark is enforced, not advisory. Fall short and the next lesson stays locked; pass and the progress record is written and the course opens up. That is what makes a completion percentage mean something to an employer looking at it later.

Angga Wika Nugraha

12

11 Graded task

A practical assignment: the brief, a link field and optional notes.

The screenshot displays the SkillLine interface for a user named Angga Wika. The main content area shows a task brief for 'TypeScript untuk Pemula'. The task requires writing three minimal TypeScript functions: one accepting a string and number, one returning a boolean, and one using an array of a specific type, all without using 'any'. Below the brief, there is a 'Link Tugas' field with a placeholder URL, a note about public access, an optional 'Catatan' field, and a 'Kumpulkan Tugas' button. The right sidebar shows the course progress for 'TypeScript untuk Pemula', listing modules and lessons. The bottom of the interface includes a footer with the SkillLine logo, a navigation menu, and social media links.

Brief: Buat minimal 3 fungsi TypeScript: satu menerima string & number, satu mengembalikan boolean, dan satu menggunakan array bertipe. Pastikan tidak ada penggunaan "any" dan semua parameter terannotasi.

Link Tugas *

Pastikan link dapat diakses publik.

Catatan (opsional)

Kumpulkan Tugas

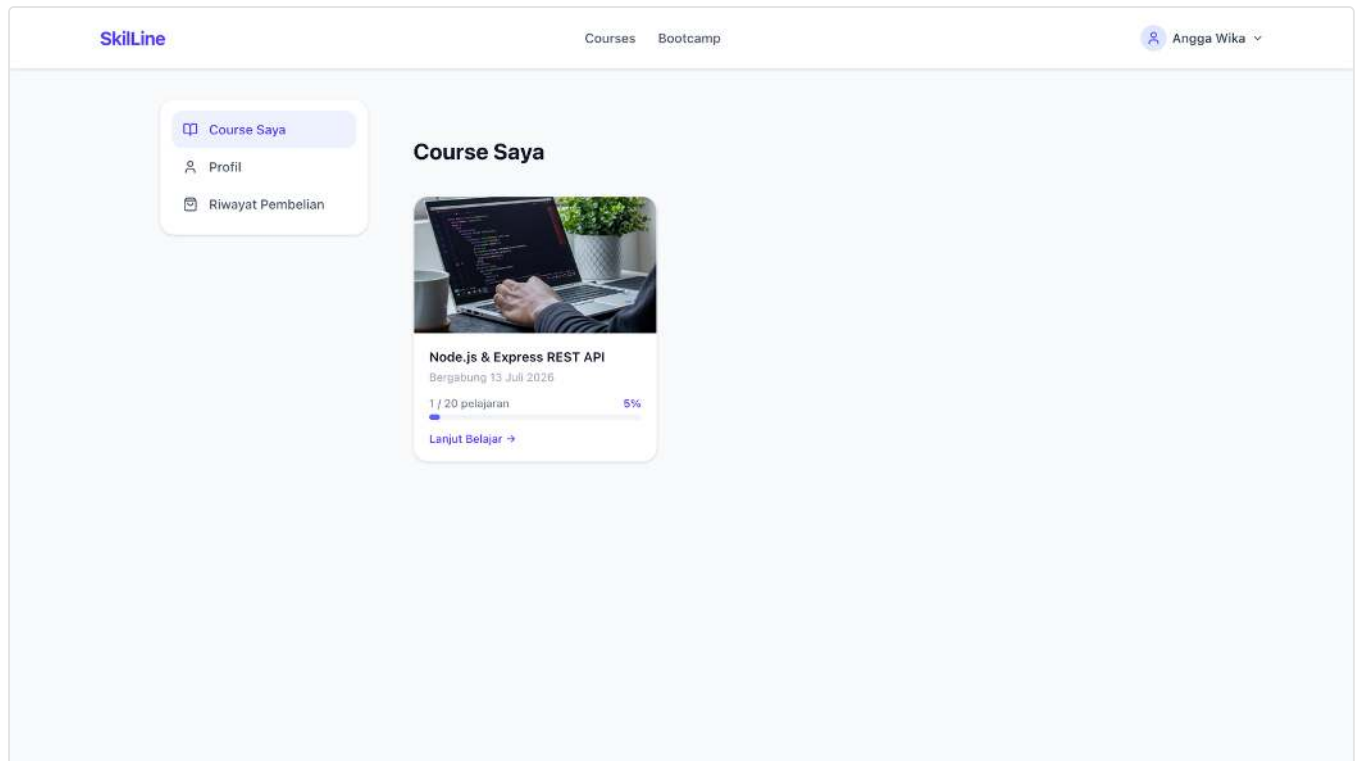
Task: Anotasi Tipe pada Fungsi Sederhana

Some things a multiple-choice question cannot test. Here the brief asks for real work — write three typed functions, annotate every parameter, use no `any` — and the student submits a link to a repository or a drive folder, with room for a note explaining their approach.

Submission does not tick the lesson off by itself. The work goes into review and the lesson carries a **Menunggu Review** badge until a human has looked at it. Slower than automatic marking, and the only honest way to grade work that has more than one correct shape.

12 My courses

Enrolled courses with progress and enrolment date.



The student's home base after signing in. Each card carries the enrolment date, lessons completed out of the total, a progress bar, and one button — **Lanjut Belajar** — which returns them to the exact lesson they stopped at.

Visible progress is not decoration on a learning platform. A bar that has moved is the main reason someone opens the site again on a tired evening.

13

Purchase history

Orders with course title, date, order reference, amount and payment status.

The screenshot shows the 'SkillLine' interface. At the top, there are navigation links for 'Courses' and 'Bootcamp', and a user profile 'Angga Wika'. On the left, a sidebar contains three menu items: 'Course Saya', 'Profil', and 'Riwayat Pembelian' (highlighted). The main content area is titled 'Riwayat Pembelian' and displays a single purchase record for the course 'Node.js & Express REST API'. The record includes a date of '15 Juli 2026', an order reference 'ORD-mr1m1aoj-suj2ji', a price of 'Rp 299.000', and a status of 'Berhasil' (Successful).

Course Title	Date	Order Reference	Amount	Payment Status
Node.js & Express REST API	15 Juli 2026	ORD-mr1m1aoj-suj2ji	Rp 299.000	Berhasil

Paid courses go through Midtrans. When the payment clears, Midtrans notifies the server directly and the server verifies that message before granting access — enrolment is never based on the browser simply saying the payment worked.

Because gateways occasionally take their time, the checkout page waits and re-checks for up to 30 seconds, and if the confirmation still has not arrived it offers a manual check rather than leaving the student staring at a spinner having just paid. Every completed order lands on this page with its reference, which is what a student sends you when something needs sorting out.

14

Profile and password

Profile details above, password change below.

The screenshot shows the 'Profil Saya' (My Profile) page in the SkillLine system. On the left is a sidebar with three menu items: 'Course Saya', 'Profil' (selected), and 'Riwayat Pembelian'. The main content area is titled 'Profil Saya' and contains two sections. The first section, 'Informasi Profil', has three input fields: 'Email' (containing 'anggawika18@gmail.com'), 'Nama' (containing 'Angga Wika'), and 'URL Foto Profil' (containing 'https://...'). Below these is a blue 'Simpan Perubahan' button. The second section, 'Ganti Password', has three input fields: 'Password Saat Ini' (with masked characters), 'Password Baru' (with a placeholder 'Minimal 8 karakter'), and 'Konfirmasi Password Baru' (with a placeholder 'Ulangi password baru'). Below these is a dark blue 'Ubah Password' button. The top navigation bar includes the 'SkillLine' logo, 'Courses', 'Bootcamp', and a user profile dropdown for 'Angga Wika'.

Name and avatar are editable; the email is fixed, because it is the verified identity the account is built on and changing it freely would undo the verification step entirely.

The password panel only appears for accounts that have a password. An account created through Google has none, and is sent through the forgot-password flow to set one — the same route described on page 6, reused rather than rebuilt.

15

Bootcamp programmes

Programme cards with enrolment status, mentors and starting price.

SkillLine

Courses Bootcamp

Angga Wika

PROGRAM INTENSIF

Bootcamp Kami

Lihat Semua →

Segera Hadir



Bootcamp Data Science & Machine Learning
Program 12 minggu dari nol hingga bisa menganalisis data dan membangun model machine learning....
 Dian Purnama
Mulai dari
Rp 6.000.000
[Lihat Detail →](#)

Buka Pendaftaran



Bootcamp Mobile Development (React Native)
Kuasai React Native dan bangun aplikasi mobile untuk iOS dan Android dalam 10 minggu. Dari setup...
 Budi Santoso, Rina Kusuma
Mulai dari
Rp 4.500.000
[Lihat Detail →](#)

Buka Pendaftaran



Bootcamp Full-Stack Web Development
Program intensif 3 bulan untuk menjadi Full-Stack Developer. Mulai dari HTML/CSS hingga membangun...
 Andi Pratama, Sari Dewi
Mulai dari
Rp 5.500.000
[Lihat Detail →](#)

SkillLine
Platform belajar teknologi terpercaya untuk karir impianmu.

NAVIGASI
[Kursus](#)
[Bootcamp](#)
[For Business](#)

IKUTI KAMI


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Bootcamps are the platform's second product and a different sale entirely — scheduled, cohort-based and priced in millions rather than hundreds of thousands.

So the cards lead with what a buyer at that price actually weighs: whether registration is open or the programme is still upcoming, who is teaching, and what it costs. The list is paginated and searchable, with filters by status.

16

Bootcamp detail

Programme description, mentor profiles, batch tabs and the session-by-session schedule.

SkillLine

CoursesBootcamp

Angga Wika [Logout](#)

Buka Pendaftaran

Bootcamp Mobile Development (React Native)

Tentang Program

Kuasai React Native dan bangun aplikasi mobile untuk iOS dan Android dalam 10 minggu. Dari setup environment hingga publish ke App Store dan Play Store dengan bimbingan mentor industri.

Mentor



Budi Santoso
Mobile Engineer @ Traveloka



Rina Kusuma
iOS Developer @ Shopee

Rp 6.000.000

Rp 4.500.000

Batch 3 · Online

Kuota

55% terisi

Daftar Sekarang

Checkout & pembayaran tersedia segera

Jadwal Batch

Batch 3Batch 4

3 Agustus 2026 – 12 Oktober 2026

Online

Kuota terisi 55%

Jadwal Sesi

 Sesi 1 — Setup React Native & Expo
Senin, 3 Agu 2026 · 13:00–16:00

 Sesi 2 — Component & Navigation
Senin, 10 Agu 2026 · 13:00–16:00

 Sesi 3 — State Management & API
Senin, 17 Agu 2026 · 13:00–16:00

 Sesi 4 — Push Notification & Camera
Senin, 24 Agu 2026 · 13:00–16:00

 Sesi 5 — Publish ke App Store & Play Store
Senin, 31 Agu 2026 · 13:00–16:00

SkillLine

Platform belajar teknologi terpercaya untuk karir impianmu.

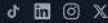
NAVIGASI

Kursus

Bootcamp

For Business

IKUTI KAMI



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Everything a prospective student asks before spending several million rupiah, on one page: what they will be able to do at the end, who is teaching and where those people work, the exact dates and times of every session, and how full the batch already is. Switching batch tabs swaps the whole schedule.

The quota bar is doing quiet sales work — 55% *terisi* says both "other people chose this" and "do not wait too long". Bootcamp checkout is the next piece to build; the card marks it plainly rather than pretending otherwise.

Angga Wika Nugraha

18

LET'S BUILD SOMETHING

Thank you for reading

This project was built end to end — database, API, interface, authentication, payment integration and assessment logic. If you need something similar, or something quite different, I would be glad to talk it through.

Name	Angga Wika Nugraha
Role	Software Engineer · Fullstack Developer
Portfolio	port-tau-azure.vercel.app
GitHub	github.com/AnggaWikaNugraha

SKILLINE

[Live demo](#) · [Source code](#) · [Project page](#)